Data Dictionary

Subscription Revenue Forecasting Dashboard

8 Tables | 3,961 Rows | Star Schema | Tableau + Hex

Overview

This data dictionary documents all 8 tables used in the Subscription Revenue Forecasting Dashboard project. It covers column names, data types, primary and foreign key relationships, field descriptions, sample values, and data quality issues identified and resolved during the analysis.

Table	Row Count	Type	Keys	Purpose
Revenue	1,000	**FACT**	PK + 4 FK	Core transaction table. Central hub of the star schema
Customers	200	**DIMENSION**	PK only	Customer master data including segment, status and region
Contracts	300	**FACT**	PK + 1 FK	Contract records with value, dates, and auto-renewal flag
Payments	1,000	**FACT**	PK + 1 FK	Payment attempt records with status (31.2% failure rate)
Products	50	**DIMENSION**	PK only	Product catalogue with category and base price
Sales_Regions	3	**DIMENSION**	PK only	Geographic region lookup: NA, EMEA, APAC
Calendar	1,096	**DIMENSION**	PK only	Date dimension covering 2022-2024 with year, quarter, month
Revenue_Forecast	12	**REFERENCE**	PK only	2025 monthly revenue predictions. Standalone — no FK joins

Fact Tables

Fact tables record business events and transactions. They contain measurable numeric values and reference dimension tables via foreign keys.

REVENUE — 1,000 rows | FACT TABLE

Central hub of the star schema. Records every revenue transaction with links to date, customer, product and region. Contains 32.2% NULL region codes (mapped to Unattributed = $1.62M).

Column Name	Data Type	Key	Description	Sample Values	Data Quality Notes
revenue_id	VARCHAR	**PK**	Unique identifier for each revenue transaction record	REV_0001	Clean
date_id	INTEGER	**FK**	Foreign key linking to Calendar table. Stored as integer (YYYYMMDD), requires conversion to DATE in SQL	20220101	Requires DATE cast
customer_id	VARCHAR	**FK**	Foreign key linking to Customers table. Identifies which customer generated this revenue	CUST_001	Clean
product_id	VARCHAR	**FK**	Foreign key linking to Products table. Identifies which product generated this revenue	PROD_01	Clean
region_code	VARCHAR	**FK**	Foreign key linking to Sales_Regions table. Indicates geographic region. NULL values treated as Unattributed	NA, EMEA, APAC	32.2% NULL — mapped to Unattributed ($1.62M)
revenue_amount	FLOAT		Monetary value of the revenue transaction in USD	1250.00	Clean

CONTRACTS — 300 rows | FACT TABLE

Records every contract between the company and its customers. Total contract portfolio value is $7.8M. 51% of records had start and end dates swapped and were corrected using CASE WHEN logic in SQL.

Column Name	Data Type	Key	Description	Sample Values	Data Quality Notes
contract_id	VARCHAR	**PK**	Unique identifier for each contract	CONT_001	Clean
customer_id	VARCHAR	**FK**	Foreign key linking to Customers table. Each customer may have multiple contracts	CUST_001	Clean
contract_start_date	DATE		Start date of the contract. NOTE: 51% of records had start/end dates swapped — fixed in clean dataset using CASE WHEN logic	2022-01-01	51% dates swapped — corrected via CASE WHEN
contract_end_date	DATE		End date of the contract. NOTE: 51% of records had start/end dates swapped — fixed in clean dataset	2023-01-01	51% dates swapped — corrected via CASE WHEN
contract_value	FLOAT		Total monetary value of the contract in USD. Avg contract value is $26,007	25000.00	Clean
auto_renew_flag	VARCHAR		Flag indicating whether the contract auto-renews. Y = auto-renews, N = manual renewal required. 47.4% are auto-renewing	Y, N	Clean — 47.4% Y, 52.6% N

PAYMENTS — 1,000 rows | FACT TABLE

Records every payment attempt. 31.2% of payments fail consistently every month — flagged as a critical infrastructure issue requiring immediate attention.

Column Name	Data Type	Key	Description	Sample Values	Data Quality Notes
payment_id	VARCHAR	**PK**	Unique identifier for each payment attempt	PAY_0001	Clean
customer_id	VARCHAR	**FK**	Foreign key linking to Customers table. Identifies which customer made the payment	CUST_001	Clean
payment_date	DATE		Date the payment was attempted	2022-03-15	Clean
payment_amount	FLOAT		Monetary value of the payment attempt in USD	1250.00	Clean
payment_status	VARCHAR		Outcome of the payment attempt. Completed = successful, Failed = declined/error, Pending = in progress. 31.2% of all payments fail every month	Completed, Failed, Pending	31.2% Failed — consistent every month (infrastructure issue)

Dimension Tables

Dimension tables provide descriptive context — who, what, where, and when. They are referenced by fact tables via foreign keys.

CUSTOMERS — 200 rows | DIMENSION TABLE

Customer master data. 34.5% of customers are churned — this rate is consistent across Enterprise, Mid-Market and SMB segments indicating a company-wide retention problem rather than a segment-specific issue.

Column Name	Data Type	Key	Description	Sample Values	Data Quality Notes
customer_id	VARCHAR	PK	Unique identifier for each customer	CUST_001	Clean
customer_name	VARCHAR		Full name of the customer or company	Acme Corp	Clean
industry	VARCHAR		Industry vertical the customer belongs to	Technology, Finance	Some NULL values present
segment	VARCHAR		Customer size segment: Enterprise, Mid-Market, or SMB	Enterprise	Clean
customer_status	VARCHAR		Current status of the customer: Active, At Risk, or Churned	Active, At Risk, Churned	Clean — 34.5% churn rate
onboarding_date	DATE		Date the customer was first onboarded to the platform	2022-03-15	Clean
region_code	VARCHAR		Geographic region of the customer. Not used as FK join in this schema but matches Sales_Regions	NA, EMEA, APAC	Some NULL values present

PRODUCTS — 50 rows | DIMENSION TABLE

Product catalogue. Revenue is split near-equally: Add-On 34.35%, Subscription 34.04%, One-Time 31.61%. The most clean table in the dataset.

Column Name	Data Type	Key	Description	Sample Values	Data Quality Notes
product_id	VARCHAR	PK	Unique identifier for each product	PROD_01	Clean
product_name	VARCHAR		Name of the product offering	Enterprise Suite	Clean
category	VARCHAR		Product category grouping: Subscription, One-Time, or Add-On. Each is approximately one third of total revenue	Subscription, One-Time, Add-On	Clean
base_price	FLOAT		Standard list price of the product in USD before any discounts	499.00	Clean

SALES_REGIONS — 3 rows | DIMENSION TABLE

Geographic region lookup table. Only 3 active regions: NA, EMEA, APAC. The region_code field has a NULL issue in the Revenue table — resolved by mapping NULLs to Unattributed in Hex SQL.

Column Name	Data Type	Key	Description	Sample Values	Data Quality Notes
region_code	VARCHAR	**PK**	Short code identifying the sales region. Only 3 active regions. NULL values in Revenue table mapped to Unattributed	NA, EMEA, APAC	NA code has NULL issue in Revenue — resolved as Unattributed
region_name	VARCHAR		Full name of the sales region	North America	Clean
country	VARCHAR		Primary country associated with the region	United States	Clean
sales_director	VARCHAR		Name of the regional sales director responsible for the territory	Jane Smith	Clean

CALENDAR — 1,096 rows | DIMENSION TABLE

Date dimension table covering 2022-2024 (3 years = 1,096 days). date_id is stored as INTEGER and must be cast to DATE when joining with Revenue. Supports time intelligence calculations in Tableau.

Column Name	Data Type	Key	Description	Sample Values	Data Quality Notes
date_id	INTEGER	**PK**	Unique integer identifier for each date. Stored as integer (YYYYMMDD). Must be cast to DATE when joining with Revenue table	20220101	Requires DATE cast for joins
cal_date	DATE		The actual calendar date	2022-01-01	Clean
year	INTEGER		Four digit calendar year	2022, 2023, 2024	Clean
quarter	INTEGER		Quarter of the year (1 through 4)	1, 2, 3, 4	Clean
month	INTEGER		Month number (1 through 12)	1, 2, 3	Clean
month_name	VARCHAR		Full name of the month	January, February	Clean
week_of_year	INTEGER		Week number within the calendar year (1 through 52)	1, 2, 52	Clean

Reference Table

Reference tables are standalone lookup tables with no foreign key relationships into the main schema.

REVENUE_FORECAST — 12 rows | REFERENCE TABLE

Contains 2025 monthly revenue predictions. This is a standalone table with no FK relationships. It was blended with historical Revenue data on a shared month axis in Tableau to create the 2025 Forecast by Month chart. Total forecast: $1,419,549.

Column Name	Data Type	Key	Description	Sample Values	Data Quality Notes
forecast_month	DATE	**PK**	The calendar month for which revenue is being forecast. Covers all 12 months of 2025	2025-01-01	Clean
predicted_revenue	FLOAT		Predicted total revenue for that month in USD. January is peak at $193,949. August is lowest at $60,534. Total 2025 forecast is $1,419,549	193949.00	Clean — standalone, no FK joins

Data Quality Summary

Three significant data quality issues were identified and resolved during this project:

Table	Issue	Severity	Root Cause	Resolution
Revenue	32.2% NULL region_code	**HIGH**	Region not captured at point of sale for $1.62M of transactions	NULLs mapped to Unattributed using ELSE clause in Hex SQL CASE WHEN
Contracts	51% swapped start/end dates	**HIGH**	Data entry error — contract end date recorded before start date	Fixed using CASE WHEN THEN ELSE to swap dates where end < start
Payments	31.2% Failed payments every month	**CRITICAL**	Consistent monthly failure rate suggests systemic payment infrastructure issue not a data issue	Flagged as business insight — not a data error. Requires payment gateway investigation
Revenue	date_id stored as INTEGER not DATE	**MEDIUM**	Calendar dimension uses integer YYYYMMDD format instead of DATE type	Cast to DATE in SQL: CAST(date_id AS DATE) for all joins

Table Relationships

This project follows a Star Schema design pattern with Revenue as the central fact table.

From Table	To Table	Join Key	Cardinality	Notes
Revenue	Customers	customer_id	Many → One	Many revenue transactions per customer
Revenue	Products	product_id	Many → One	Many revenue transactions per product
Revenue	Sales_Regions	region_code	Many → One	32.2% NULL — mapped to Unattributed
Revenue	Calendar	date_id	Many → One	date_id must be cast from INTEGER to DATE
Customers	Contracts	customer_id	One → Many	One customer can have multiple contracts
Customers	Payments	customer_id	One → Many	One customer can have multiple payment attempts
Revenue_Forecast	Standalone	None	N/A	No FK joins. Blended with Revenue on shared month axis in Tableau only